

PAPER_1715_SHEATH_PHI_TE_2_84

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PAPER_1715 — Plasma Sheath Potential $\phi_{sh}/T_e = \phi_{sh}/T_e = K + \Phi - F + F^2 \cdot K + F^3 \approx 2.84$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Plasma Physics

Date: June 18, 2026

Status: CLOSED — 0.05% closure (PAPER_1196 plasma fusion proof set)

Observation

PAPER_1196_S422 (PAPER_1196 Plasma Fusion Unified Proof Set) gives:

``

$$\phi_{sh}/T_e = K + \Phi - F + F^2 \cdot K + F^3 \approx 2.84$$

``

Status: 0.05%.

UQFF Closed Identity

``

$$\phi_{sh}/T_e = K + \Phi - F + F^2 \cdot K + F^3 \approx 2.84 \text{ } 0.05\%$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

Reference

- Source: PAPER_1196_S422

- Related: PAPER_1696-1705 (tier-28); PAPER_1686-1695 (tier-27)

- Calculator dispatch: `calculate_paradox({"paradox": "sheath_phi_te_2_84"})`

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